Quyidagi differensial tenglamalarning chap tomonlari to‘liq differensialdan iborat ekanligi tekshirilsin va tenglamalar yechilsin:

1) $(e^x$+y+siny)dx + $(e^y$+x+xcosy)dy =0

Bu yerda M(x,y) = e^x+y+siny, N(x,y) = e^y+x+xcosy

$\frac{\partial M}{\partial y}$ = 1+cosy, $\frac{\partial N}{\partial x}$ = 1+cosy,

$\frac{\partial u}{\partial x}$ = M(x,y) bo’lganligi sababli ,

$\frac{\partial u}{\partial x} = e^x$+y+siny.

Bu tenglikni x bo’yicha integrallaymiz:

u = $e^x + \varphi(y)$

bundan:

$\frac{\partial u}{\partial y}$ = N(x,y) ekanligini hisobga olsak,

$\frac{\partial u}{\partial y} = \varphi'(y)$ dan :

$\varphi'(y)$ = e^y+x+xcosy yoki $\frac{\partial \varphi}{\partial y}$ = e^y+x+xcosy

$\varphi(y) = e^y$- x siny + C

Javob: u = $e^x + e^y$- x siny + C

Yoki $e^x + e^y$- x siny = C

2) (x+y-1)dx + (e^y+x)dy = 0

Bu yerda M(x,y)= x+y-1, N(x,y)= e^y+x.

$\frac{\partial M}{\partial y}$ = 1, $\frac{\partial N}{\partial x}$ = 1, ya'ni $\frac{\partial M}{\partial y}$ = $\frac{\partial N}{\partial x}$

$\frac{\partial u}{\partial x}$ = M(x,y) bo'lganligi sababli:

$\frac{\partial u}{\partial x}$ = x+y-1

Bu tenglikni x bo'yicha integrallaymiz:

u = $\frac{x^2}{2}$ –x + $\varphi(y)$.

Bundan $\frac{\partial u}{\partial y}$ =N(x,y) ekanligini hisobga olsak,

$\frac{\partial u}{\partial y}$ = $\varphi'(y)$ = e^y+x.

$\varphi'(y)$ = e^y+x bundan $\varphi(y)$= e^y + xy + C

Javob: u = $\frac{x^2}{2}$ –x + e^y + xy + C

3) e^x(xcosy – ysiny)dy + e^x(xsiny – ycosy)dx = 0

Bu yerda M(x,y)= e^x(xcosy – ysiny), N(x,y)= e^x(xsiny – ycosy)

$\frac{\partial M}{\partial y}$ = e^xxsiny - e^xsiny - e^xycosy,

$\frac{\partial N}{\partial x}$ = e^xxsiny - e^xsiny - e^xycosy , $\frac{\partial M}{\partial y}$ = $\frac{\partial N}{\partial x}$.

$\frac{\partial u}{\partial x}$ = M(x,y) bo'lganligi sababli

$\frac{\partial u}{\partial x}$ = e^x(xcosy – ysiny)

Bu tenglikni x bo'yicha integrallaymiz:

u = e^xxcosy - e^xcosy - e^xysiny + $\varphi(y)$

Bundan $\frac{\partial u}{\partial y}$ =N(x,y) ekanligini hisobga olsak,

$\frac{\partial u}{\partial y}$ = e^xxsiny - e^xsiny - e^xsiny - $e^x y$cosy + $\varphi'(y)$

$\frac{\partial u}{\partial y}$ = e^xxsiny - 2e^xsiny - $e^x y$cosy + $\varphi'(y)$

e^xxsiny – e^xycosy = e^xxsiny - 2e^xsiny - $e^x y$cosy + $\varphi'(y)$

$\varphi'(y)$ = - 2e^xsiny

$\varphi(y)$ = - 2e^xcosy

u = e^xxcosy - e^xcosy - e^xysiny - 2e^xcosy

u = e^xxcosy - 3e^xcosy - e^xysiny

4) $2xydx + (x^2 - y^2)dy = 0$

Bu yerda M(x,y) = 2xy , N(x,y)= $x^2 - y^2$;

$\frac{\partial M}{\partial y} = 2x$, $\frac{\partial N}{\partial x} = 2x$, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

$\frac{\partial u}{\partial x}$ = M(x,y) bo'lganligi sababli

$\frac{\partial u}{\partial x} = 2xy$

Bu tenglikni x bo'yicha integrallaymiz:

$u = y\,x^2 + \varphi(y)$

Bundan $\frac{\partial u}{\partial y}$ =N(x,y) ekanligini hisobga olsak,

$\frac{\partial u}{\partial y} = x^2 + \varphi'(y)$

$x^2 - y^2 = x^2 + \varphi'(y)$

$\varphi'(y) = -y^2$ bundan $\varphi(y) = -y^3 + C$

$u = y\,x^2 - y^3 + C$

5) $(2 - 9xy^2)xdx + (4y^2 - 6x^3)ydy = 0$

Bu yerda $M(x,y) = 2 - 9xy^2$, $N(x,y) = 4y^2 - 6x^3$

$\frac{\partial M}{\partial y} = -18xy$, $\frac{\partial N}{\partial x} = -18xy$, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

$\frac{\partial u}{\partial x} = M(x,y)$ bo'lganligi sababli

$\frac{\partial u}{\partial x} = 2 - 9xy^2$,

Bu tenglikni x bo'yicha integrallaymiz:

$u = 2x - \frac{9x^2y^2}{2} + \varphi(y)$

Bundan $\frac{\partial u}{\partial y} = N(x,y)$ ekanligini hisobga olsak,

$\frac{\partial u}{\partial y} = -9x^2y + \varphi'(y)$

$4y^2 - 6x^3 = -9x^2y + \varphi'(y)$

$\varphi'(y) = 4y^2 - 6x^3 + 9x^2y$

$\varphi(y) = \frac{4}{3}y^3 + \frac{9}{2}x^2y^2 + C$

$u = 2x - \frac{9x^2y^2}{2} + \frac{4}{3}y^3 + \frac{9}{2}x^2y^2 + C$

Javob: $u = 2x + \frac{4}{3}y^3 + C$

Quyidagi differensial tenglamalarning integrallovchi ko‘paytuvchilari topilsin va tenglamalar echilsin.

1) $(x^2 - y)dx + xdy = 0$

Bu yerda M(x,y) = x^2 – y , N(x,y) = x ,

$\frac{\partial M}{\partial y} = -1$, $\frac{\partial N}{\partial x} = 1$, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$;

Demak, tenglamaning chap tomoni biror funksiyaning to‘la differensiali emas. Bu tenglamaning faqat y ga bog‘liq bo‘lgan integrallovchi ko‘paytuvchisi bormi degan masalani qaraymiz.

$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{1+1}{x^2 - y} = \frac{2}{x^2 - y}$ bunda $\varphi_1(y)$ funksiya to’la y ga bog’liq emas.

$\varphi(x) = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{-1-1}{x} = -\frac{2}{x}$

$\ln\mu = \int \varphi(x)dx$ ekanligidan $\ln\mu = \int -\frac{2}{x}dx$,

$\ln\mu = -2\ln x$,

$\mu = \frac{1}{x^2}$ integrallovchi ko’paytuvchini ifodaga ko’paytiramiz:

μ M(x,y)dx + N(x,y)dy = 0

Berilgan tenglamaning ga ko‘paytirganda keyin:

$\frac{1}{x^2}(x^2 - y)dx + \frac{1}{x^2}xdy = 0$

Bundan M(x,y) = 1- $\frac{y}{x^2}$, N(x,y) = $\frac{1}{x}$;

$\frac{\partial M}{\partial y} = -\frac{1}{x^2}$, $\frac{\partial M}{\partial y} = -\frac{1}{x^2}$, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Endi oddiy $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ hol uchun tenglamani ishlab qo’yamiz:

$\frac{\partial u}{\partial x} = 1 - \frac{y}{x^2}$; u = x - $\frac{y}{x} + \varphi(y)$

Bundan $\frac{\partial u}{\partial y}$ =N(x,y) ekanligini hisobga olsak,

$\frac{\partial u}{\partial y} = -\frac{1}{x} + \varphi'(y) = \frac{1}{x}$

$\varphi'(y) = \frac{2}{x}$; $\varphi(y) = 2\ln x + C$;

Javob: $u = x - \frac{y}{x} + 2\ln x + C$;

2) $y^2dx + (yx - 1)dy = 0$

Bu yerda $M(x,y) = y^2$, $N(x,y) = yx - 1$,

$\frac{\partial M}{\partial y} = 2y$, $\frac{\partial N}{\partial x} = y$, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$;

Demak, tenglamaning chap tomoni biror funksiyaning to‘la differensiali emas. Bu tenglamaning faqat y ga bog‘liq bo‘lgan integrallovchi ko‘paytuvchisi bormi degan masalani qaraymiz.

$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{y-2y}{y^2} = -\frac{1}{y}$ bunda $\varphi_1(y)$ funksiya to’la y ga bog’liq .

$\ln\mu = \int \varphi_1(y)\, dy$ ekanligidan $\ln\mu = \int -\frac{1}{y}\, dy$, $\ln\mu = -\ln y$;

$\mu = \frac{1}{y}$ integrallovchi ko’paytuvchini ifodaga ko’paytiramiz:

$\mu\, M(x,y)dx + N(x,y)dy = 0$

Berilgan tenglamaning ga ko‘paytirganda keyin:

$\frac{1}{y}\, y^2dx + \frac{1}{y}(yx - 1)\, dy = 0$

Bundan $M(x,y) = y$, $N(x,y) = x - \frac{1}{y}$;

$\frac{\partial M}{\partial y} = 1$, $\frac{\partial M}{\partial y} = 1$, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Endi oddiy $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ hol uchun tenglamani ishlab qo’yamiz:

$\frac{\partial u}{\partial x} = y$; $u = yx + \varphi(y)$

Bundan $\frac{\partial u}{\partial y} = N(x,y)$ ekanligini hisobga olsak,

$\frac{\partial u}{\partial y} = x + \varphi'(y) = x - \frac{1}{y}$,

$\varphi'(y) = -\frac{1}{y}$; $\varphi(y) = -\ln y + C$;

Javob: $u = yx - \ln y + C$;

3) $(e^{2x} - y^2)dx + ydy = 0$

Bu yerda M(x,y) = $e^{2x} - y^2$, N(x,y) = y ,

$\frac{\partial M}{\partial y} = -2y$, $\frac{\partial N}{\partial x} = 0$, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$;

Demak, tenglamaning chap tomoni biror funksiyaning to‘la differensiali emas. Bu tenglamaning faqat y ga bog‘liq bo‘lgan integrallovchi ko‘paytuvchisi bormi degan masalani qaraymiz.

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{0+2y}{e^{2x} - y^2} = \frac{2y}{e^{2x} - y^2}$$

bunda $\varphi_1(y)$ funksiya to’la y ga bog’liq.

$$\varphi(x) = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{-2y-0}{y} = -2$$

$\ln\mu = \int \varphi(x)dx$ ekanligidan $\ln\mu = \int -2\, dx$,

$\ln\mu = -2yx$,

$\mu =$ integrallovchi ko’paytuvchini ifodaga ko’paytiramiz:

μ M(x,y)dx + N(x,y)dy = 0

Berilgan tenglamaning μ ga ko‘paytirganda keyin:

$-2xy(e^{2x} - y^2)dx - 2xy^2\, dy = 0$

Bundan M(x,y) = $2xy^3$, N(x,y) = ;

4) $xy^2(xy` + y) = 1$

$x^2y^2y` = 1 - xy^3$

$x^2y^2\, dy + (xy^3 - 1)dx = 0$

Bu yerda M(x,y) = x^2y^2 , N(x,y) =$xy^3 - 1$,

$\frac{\partial M}{\partial y} = 2x^2y$, $\frac{\partial N}{\partial x} = y^3$, $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$;

Demak, tenglamaning chap tomoni biror funksiyaning to‘la differensiali emas. Bu tenglamaning faqat y ga bog‘liq bo‘lgan integrallovchi ko‘paytuvchisi bormi degan masalani qaraymiz.

$$\frac{\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}}{M}=\frac{y^3-2x^2\mathrm{y}}{x^2y^2}=\frac{y}{x^2}-\frac{2}{y}$$

bunda $\varphi_1(y)$ funksiya to’la y ga bog’liq emas.

$$\varphi(x)=\frac{\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}}{N}=\frac{2x^2\mathrm{y}-y^3}{xy^3-1}$$